LAB ASSIGNMENT-02

Exploring packet sniffer and packet Analyzer :-

Question-01: What are the different protocals you observe at the following layers of the protocol stack?

Solution :- (A) Application layer

- HTTP
- HTTPS / TLS
- DNS

(B) Transport layer

- TCP
- UDP

(C) Network layer

- IPv4

Question-2: What is the total amount of data being recieved for the following two cases?

(A) When you access http://neverssl.com/

(B) When you access https://www.cornell.edu.

Solution :-

(A) While accessing http://neverssl.com/ the total data recie-ved are 2273 bytes.

(B) while accessing https://www.cornell.edu the total data received are 17,234 bytes.

Question-03:- Answer the following when you access the site http://neverssl.com/.

(A) How many HTTP GET request do you observe? List down the GET request.

Solution:- There are 3 HTTP GET requests

- GET/online HTTP/1.1
- GET/online HTTP/1.1
- GET/favicon.ico HTTP/1.1

(B) What version of HTTP the server is running.

Solution:- HTTP/1.1

(C) For each of the HTTP GET request find out:

i) The total number of TCP segments being received

Solution:- 2 TCP segments

ii) The total amount of data being recieved in the corresponding HTTP response message.

Solution:- 2273 bytes of data.

(D) Identify TCP three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each.

Solution:-

Packet	Sequence Number
SYN	917965950
SYN-ACK	1837812572 (ACK = 917965951)
ACK	917965951 (ACK = 1837812573)